Adaptation Needs in Robotic Systems: Behavior Trees and Beyond

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***Abstract*—Robotic systems increasingly operate in dynamic, uncertain, and open-ended environments, where design-time assumptions may no longer hold during operation. In such settings, robots may need to adapt their behavior, knowledge, configuration, or mission interpretation to remain effective and safe. Thus, the robot's control architecture must support adaptation either directly or through integration with external mechanisms. Behavior Trees (BTs) are widely used to implement robotics control architectures because they provide a modular, readable, and reactive representation of task-level behavior. This raises a central question: are BTs sufficient to meet the adaptation needs of modern robotic systems?**

This paper investigates this question through a survey. First, we derive a classification of robotic adaptation needs from literature on uncertainty, self-adaptive systems, and robotics, organizing them into six categories: Knowledge, Perception, Actuation, System, Mission, and Environment. Then, we analyze the features of classical BTs and learn their limitations with respect to these needs. Finally, based on 31 primary studies, we characterize BT-based approaches for adaptation, including generation, extension, evolution, refinement, and hybrid techniques.

Our analysis shows that the classical BTs' support for modularity, flexibility, and reactivity is insufficient on its own for tackling adaptation needs that involve runtime restructuring, reasoning under uncertainty, mission reinterpretation, learning, or integration with external knowledge and planning mechanisms. Beyond this observation, our findings show how classical BT features and limitations relate to different adaptation needs, and to what extent different BT-based techniques are suitable for addressing each family of adaptation needs. Furthermore, the resulting classification and characterization provide guidance on when classical BTs are appropriate, when enhanced BT mechanisms are required, and which open challenges remain for adaptive robotic control architectures.

***Index Terms*—Behavior Trees, Adaptation, Uncertainty, Robotic systems.**

I. Introduction

Robotic systems increasingly operate in dynamic, partially observable, and open-ended environments, where environmental changes, uncertainty, and evolving requirements affect their ability to perceive the world, act, and satisfy mission objectives [1]–[8]. In such settings, assumptions made at design time may no longer hold during operation: sensors may provide noisy or conflicting data, actuators may behave non-deterministically, resources may fail or change, missions may evolve, and humans or other agents may interact with the system in an unpredictable way and modify the environment.

The violation of design-time premises creates a *need to adapt* the robotic system's behavior, knowledge, configuration, or mission interpretation in order to maintain or improve mission achievement in response to changes [1]. Throughout this paper, we refer to the conditions that trigger adaptation as *adaptation needs*. To cope with such needs, robotic systems require control architectures that allow for reacting to changing conditions at runtime while maintaining correct task execution. In this context, Behavior Trees (BTs) have become a prominent representation for robot behavior because they support modularity, hierarchy, reuse, and enable reactive execution, while offering an intuitive structure for task-level control [9]–[12]. Recent empirical work further shows that BTs are used in a variety of robotics applications and reports that, compared with traditional behavior modeling formalisms such as state machines, they are better suited for developing modular, reactive, and adaptable robot behaviors [13], [14]. These properties make BTs attractive for structuring robot behavior in systems that must react to changing execution conditions.

However, the same scenarios that motivate the use of BTs also expose their limitations. Classical BTs typically encode predefined control flows, condition checks, and recovery strategies. Consequently, the complexity of the overall system architecture tends to increase when adaptation requires runtime restructuring, explicit reasoning under uncertainty, mission reinterpretation, learning, or integration with planning and knowledge-management mechanisms. The growing body of work that extends BTs with planning, active inference, adaptive nodes, architectural adaptation, and uncertainty-aware mechanisms suggests that classical BTs alone may be insufficient for several forms of robotic adaptation [15]–[19].

Existing surveys have examined BT concepts, robotics applications, formal and control-theoretic properties, and their integrations with reinforcement learning and learning from demonstration [10]–[14], [20]. Nevertheless, these works do not systematically start from the adaptation needs of real-world robotic systems and analyze whether and how such needs can be addressed by classical BT features, linguistic enhancements of BTs, or online evolutionary algorithms. Consequently, it remains unclear which adaptation needs are naturally tackled by classical BTs,

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which require extensions or complementary mechanisms, and remain insufficiently addressed in practice and in the scientific literature.

This paper addresses this gap by investigating the following research questions:

- **RQ1**: What adaptation needs characterize robotic systems?
- **RQ2**: To what extent can classical BT features address these adaptation needs, and what limitations emerge?
- **RQ3**: To what extent do approaches and techniques that have been proposed to enhance BTs overcome the limitations of classical BTs in addressing these adaptation needs, and what new limitations emerge?

To answer these questions, we first derived a classification of adaptation needs by combining uncertainty classifications from self-adaptive systems with robotics-specific literature. Second, we analyzed BT adaptation or evolution ~~studies~~ → approaches **Gianluca** *Not sure about the right word to use here. Next we say "those approaches" so it looks like we already checked the approaches. However, RQ2 focuses on the classical BTs...* in robotic systems proposed in the literature to identify the features and limitations of classical BTs with respect to those needs. Third, we categorized those approaches to understand how they compensate for the identified limitations. Finally, we synthesized remaining challenges and identified the capabilities that future behavior modeling formalisms should provide to better support adaptation needs in robotic systems. The literature-driven analysis is based on a corpus of 31 primary studies on BTs, adaptation, evolution, and robotics.

The main contributions of this paper are:

- A classification of adaptation needs for robotic systems, organized into six categories: Knowledge, Perception, Actuation, System, Mission, and Environment.
- An analysis of the features and limitations of classical BTs with respect to these adaptation needs.
- A mapping between robotic adaptation needs and existing BT-based approaches.
- A categorization of approaches used to enhance BTs for adaptation, including BT generation, extension, evolution, refinement, and hybrid approaches.
- A discussion of open research gaps and future directions for adaptive robotic control architectures beyond classical BTs.

To validate our findings, we conducted a questionnaire-based evaluation with robotics and BT practitioners. We collected feedback on the identified adaptation needs, the suitability of BT-based approaches to address adaptation needs, and the limitations of existing solutions. The responses were analyzed to assess the agreement between our literature-driven classification and practitioners' perspectives. These provided empirical evidence to support the identified challenges and research directions.

Our results show that, although classical BTs provide fundamental software engineering capabilities for adaptive robotic systems– such as modularity, hierarchical decomposition, reusability, and reactive execution– they are insufficient to fully support complex adaptation needs arising from uncertainty, changing missions, and dynamic interactions with humans and environments. The analysis reveals that current BT extensions, evolution, generation, and refinement approaches introduce complementary capabilities, but no single approach provides a complete solution. These findings highlight the need for future behavior modeling formalisms that integrate adaptation mechanisms as first-class concepts rather than treating adaptation as an external capability. This work contributes to software engineering research on self-adaptive systems. It uses robotics as a demanding application domain to characterize the adaptation capabilities that behavioral control architectures need. These capabilities allow software-intensive systems to remain effective under changing operational conditions.

The remainder of this paper is organized as follows. Section II introduces the background concepts. Section III describes the research methodology. Section IV presents the classification of adaptation needs for robotic systems, answering RQ1. Section V analyzes the features and limitations of classical BTs, answering RQ2. Section VI categorizes adaptive BT approaches, answering RQ3. Section VII discusses the findings and open challenges. Section VIII positions the paper with respect to related work. Finally, Section IX concludes the paper.

II. Background

This section introduces the concepts needed to understand the analysis conducted in the paper. We first recall the basic structure of BTs, since they are the object of our review. We then clarify how we use the terms *generation*, *extension*, *evolution*, and *refinement*, because these terms are used to classify the approaches analyzed in RQ3. Finally, we introduce the notion of *adaptation need*, which is central to the taxonomy presented in RQ1. A complete list of definitions adopted in this paper is reported in Appendix A.

A. Behavior Trees (BTs)

BTs are hierarchical control structures used to organize the behavior of autonomous agents. A BT represents behavior as a tree of modular nodes. Internal nodes define the control flow, while leaf nodes represent actions or conditions. Common control-flow nodes include sequence, fallback, parallel, and decorator nodes, which determine how actions and conditions are selected and executed [9].

BTs are widely used in robotics because they provide a readable and modular representation of task-level behavior. Their hierarchical structure allows complex missions to be decomposed into smaller sub-behaviors, while their execution semantics support reactivity through repeated ticking and condition evaluation. These properties make BTs attractive for specifying robot behavior in systems that need to react to changing execution conditions.

However, classical BTs provide only limited support for advanced adaptation requirements. In many cases, the adaptation logic must already be encoded in the tree through conditions, fallback branches, or recovery subtrees. When adaptation requires runtime restructuring, reasoning under uncertainty, mission reinterpretation, quality-aware decision making, or integration with planning and learning, classical BTs may need to be extended or combined with additional mechanisms. This motivates the analysis conducted in Sections V and VI.

B. Creating and Modifying Behavior Trees

In this paper, we distinguish four ways in which BTs can be created or modified for supporting adaptation: *generation*, *extension*, *evolution*, and *refinement*. This distinction is important because different forms of BT adaptation operate on different inputs and produce different kinds of changes.

BT generation refers to processes that produce a BT from a higher-level specification, such as a task model, natural-language instruction, demonstration, plan, or user intent. In generation, there is no initial BT that is taken as the behavior to be modified, although existing BTs, fragments, templates, libraries, or design patterns may be used as prior knowledge to guide the generation process.

BT extension refers to processes that change the BT formalism itself. This may include adding new types of nodes, modifying the semantics of existing nodes, introducing decorators, or integrating BT execution with additional reasoning mechanisms. Extension, therefore, changes what can be expressed within the BT language or how BT execution is interpreted.

BT evolution refers to processes that start from an initial, often minimal or partial, BT and progressively modify its structure to obtain a tree that encodes more complex or improved behavior. Evolution is commonly associated with search-based or optimization-based techniques, such as genetic programming or grammatical evolution.

BT refinement refers to processes that take an existing BT as input and update it to correct, improve, or modify its behavior without changing the underlying BT formalism. Refinement may modify nodes, parameters, subtrees, or execution conditions, and it may occur during design time or at runtime.

These four concepts are used in RQ3 to categorize how the literature enhances BTs for adaptation. Some approaches combine more than one form of change. For example, an approach may first generate an initial BT from a demonstration and then refine it through interaction or optimization.

C. Adaptation Needs in Robotic Systems

We use the term *adaptation need* to denote a condition indicating that a robotic system should adapt its current behavior, knowledge, configuration, or mission interpretation to maintain or improve mission achievement in response to observed or anticipated changes in the system, its environment, available knowledge, perception, or objectives [21], [22].

Adaptation needs may arise from uncertainty, failures, changing resources, mission updates, unreliable perception, non-deterministic actuation, or interactions with humans and other agents. In a nutshell, adaptation needs are the reason leading a robotic system to perform adaptation.

In this paper, adaptation needs are used as the organizing lens for analyzing BTs. Rather than asking only whether BTs are modular, readable, or reactive, we ask whether these properties are sufficient to address the situations in which robotic systems need to adapt. Section IV derives a classification of adaptation needs organized into six categories: *Knowledge*, *Perception*, *Actuation*, *System*, *Mission*, and *Environment*. These categories capture the main parts of a robotic system from which adaptation needs may emerge.

III. Methodology

This research aims not only to review existing literature but also to incrementally construct and validate an artifact that supports the analysis of adaptation needs in robotic systems and their relationship with BTs. In this work, the artifact consists of a set of structured classifications that progressively capture: (i) adaptation needs in robotic systems, (ii) the extent to which such needs are addressed by BT-based approaches, and (iii) the categories of BT-based adaptation or evolution approaches employed in robotics. The research process is organized into three iterative cycles. In each cycle, first, we identified and framed the problem addressed in the cycle, then constructed or refined the corresponding part of the artifact. At the end of each cycle, we validated the outcome of the cycle either by co-authors' or experts' validation to determine whether further refinement is needed. In addition to internal validation by the co-authors, we validated the final artifact through three empirical activities: (i) a targeted author validation in which we asked authors of the primary studies to assess the appropriateness of our mapping of their techniques to the four BT adaptation classes; (ii) a questionnaire with researchers and practitioners knowledgeable in robotics and BTs; and (iii) semi-structured interviews with domain experts from both industry and academia. Figure 1 provides an overview of the methodology adopted in this work.

The cycles are cumulative. Cycle 1 focuses on identifying adaptation needs of robotic systems. Cycle 2 investigates how such needs are reflected in the literature on BTs and robotic adaptation. Cycle 3 categorizes the approaches used to support robotic missions by adapting or evolving BTs. Finally, we identified research gaps in adaptation needs in BT-based robotic systems, which were gradually obtained through the cycles.

A. Cycle 1: Problem Exploration and Identification of Adaptation Needs

The first cycle addresses RQ1 by identifying and organizing the major adaptation needs that arise in robotic

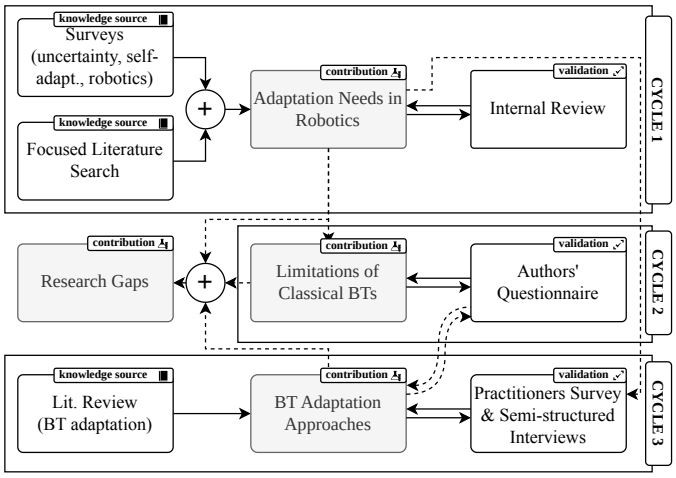

Fig. 1. Research design for identifying adaptation needs, assessing classical BTs, characterizing BT-based adaptation approaches, and deriving research gaps.

systems. The goal of this cycle is to build a robotics-specific classification of adaptation needs grounded in both self-adaptive systems research and robotics literature.

In the first phase, we investigated how uncertainty and adaptation have been conceptualized in self-adaptive systems and robotics. We started from established literature on uncertainty in self-adaptive systems, including studies on uncertainty dimensions and descriptions [23], [24], uncertainty taxonomies [6], [25], and classification studies in self-adaptive systems [5], [26], [27]. We also considered robotics-oriented sources, including studies on robotic uncertainty and modeling [8] and the Multi-Annual Roadmap for Robotics [7]. The rationale behind this step is that uncertainty is one of the main drivers of adaptation. Robotic systems operate in dynamic, partially observable, and often unpredictable environments. As a result, adaptation needs may arise from several sources, including limitations in knowledge, perception, actuation, system capabilities, mission specification, and environmental conditions.

In the next phase, we constructed an initial classification of adaptation needs for robotic systems. We first extracted uncertainty-related concepts and descriptions from the self-adaptive systems literature. We then interpreted these concepts in the context of robotics by identifying the corresponding robotic-system elements affected by uncertainty.

To ground the classification in robotics literature, we conducted targeted searches in Google Scholar, Scopus, and ACM Digital Library. For each candidate adaptation need, we defined specific search strings combining general terms, such as *"robotic"*, *"adaptation"*, and *"uncertainty"*, with terms related to the specific adaptation category under analysis. All search strings are available in the dedicated section of the replication package [28]. The purpose of these searches was to identify concrete examples, definitions, and evidence from the robotics literature that could support, refine, or challenge the initial classification.

We included studies that met at least one of the following criteria: (i) they discuss an adaptation need or adaptation strategy in the robotic domain; or (ii) they discuss adaptation in autonomous or self-adaptive systems while providing examples or applications in robotics. Through this process, we consolidated the adaptation needs into six main categories: *Knowledge*, *Perception*, *Actuation*, *System*, *Mission*, and *Environment*.

B. Cycle 2: Literature-Driven Analysis of BT Limitations and Adaptation Needs

The second cycle addresses RQ2 by investigating how the adaptation needs identified in Cycle 1 are reflected in the literature on BTs for robotic systems. The goal of this cycle is to understand the extent to which classic BTs are sufficient to address such needs, and why researchers have proposed extensions, adaptations, or alternative BT-based mechanisms.

In the first phase, we identified studies that use, adapt, evolve, refine, or extend BTs in robotic systems. To obtain relevant keywords, we first analyzed terminology used in studies aligned with our research questions. The most recurring and relevant terms included *"behavior tree"*, *"behaviour tree"*, *"robotics"*, *"adaptation"*, and *"evolution"*. After discussion among the co-authors, we selected the following search string:

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("behavior tree" OR "behaviour tree")
AND (adapt* OR evol*)
AND robot*
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The search was conducted through SciVal, restricted to the robotics domain. The retrieved studies were screened to identify papers relevant to the objectives of this research. In particular, we selected studies that present, discuss, or evaluate adaptation mechanisms involving BTs in robotic systems. This process resulted in 31 primary studies, which are reported in the replication package [28].

In the next phase, we analyzed the selected studies from three perspectives. First, we extracted the BT features that make BTs suitable for robotic control, such as modularity, reactivity, readability, and compositionality. Second, we identified limitations of classical BTs when they are used to address adaptation needs in robotic systems. Third, we mapped the adaptation needs from Cycle 1 to the selected BT-related studies.

This mapping allowed us to examine which adaptation needs are explicitly addressed in the literature and which remain weakly covered. It also helped us identify why classic BTs alone may be insufficient for managing some robotic adaptation needs. For example, some needs require runtime modification of the tree, integration with learning-based components, explicit handling of uncertainty, or mechanisms for updating knowledge and goals during execution.

In addition, we extracted technical information about the proposed solutions in the selected studies. This included the techniques used to support adaptation, the type of robotic system considered, the adaptation triggers, the adaptation goals, and the level of human involvement.